

TikTok Video Performance Analysis

Group 4

Table of contents

1	1. Introduction	2
2	2. Data Inspection	2
2.1	2.1 Loading the data	2
2.2	2.2 Video-view dataset overview	3
2.3	2.3 Impression dataset overview	6
2.4	2.4 Watch-event dataset overview	8
3	3. Descriptive Analysis	11
3.1	3.1 Overall video performance	11
3.2	3.2 Watch share	12
3.3	3.3 Most viewed video	13
3.4	3.4 Distribution of video length	13
3.5	3.5 Videos with high watch rates	14
3.6	3.6 Top 10 videos by watch rate	15
3.7	3.7 Performance by creator	16
4	4. Watch-Event Analysis	17
4.1	4.1 Aggregating watch events	17
4.2	4.2 Combining impression and watch data	17
4.3	4.3 Distribution of watch time	19
5	5. Regression Analysis	19
5.1	5.1 Simple regression	20
5.2	5.2 Multiple regression	20
5.3	5.3 User fixed effects	22
6	6. Conclusions	39
6.0.1	6.0.1 What did the regression analysis test?	40
6.0.2	6.0.2 Model 1: Recommendation score only	40

6.0.3	Model 2: Adding other recommendation factors	40
6.0.4	Model 3: Also accounting for differences between users	41
6.0.5	What does this mean overall?	41
7	7. Limitations	42
8	8. Overall Conclusion	42

1 1. Introduction

This project analyses TikTok video performance to better understand differences in reach and watch behaviour. The analysis begins with an exploration of the data, focusing on video and creator performance, including impressions, watch rates, and watch time.

The analysis then uses regression models to examine whether characteristics of an impression are associated with the amount of time users spend watching the content. Together, these analyses provide insight into patterns in the data and the factors that are associated with user engagement.

2 2. Data Inspection

2.1 2.1 Loading the data

The analysis uses the cleaned datasets stored in the project's processed data folder, as well as the original video-view dataset.

```
library(here)
library(tidyverse)
library(DBI)
library(RSQLite)

# Video-view data from SQLite
con <- dbConnect(
  SQLite(),
  here("data", "raw", "tiktok_students.sqlite")
)

views <- dbGetQuery(
  con,
  "SELECT * FROM video_view"
)
```

```
dbDisconnect(con)

# Cleaned impression data
impressions <- read.csv(
  here("data", "processed", "impressions_clean.csv")
)

# Cleaned watch-event data
watch_events <- read.csv(
  here("data", "processed", "watch_events_clean.csv")
)
```

2.2 2.2 Video-view dataset overview

The first step is to inspect the structure and quality of the video-view dataset.

```
nrow(views)
```

```
[1] 50000
```

```
str(views)
```

```
'data.frame': 50000 obs. of 16 variables:
 $ video_id      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ creator_id    : int  27 346 212 410 279 393 390 255 136 69 ...
 $ video_length_sec : num  33 27 18 10 30 17 36 17 25 29 ...
 $ published_at   : int  1775374080 1758594709 1759578576 1764529326 17851581...
 $ video_url      : chr   "https://www.tiktok.com/@sophiadavis/video/9000000000...
 $ impressions_n  : int  25 50 61 44 0 35 13 4 16 99 ...
 $ watched_n     : int  19 34 42 32 0 30 11 4 12 67 ...
 $ watch_rate     : num  0.76 0.68 0.689 0.727 NA ...
 $ avg_watch_seconds_when_watched: num  15.68 14.85 10.29 5.34 NA ...
 $ total_watch_seconds : num  298 505 432 171 0 ...
 $ avg_watch_share : num  0.361 0.374 0.393 0.389 NA ...
 $ distinct_watchers_n : int  19 33 39 32 0 30 11 4 11 64 ...
 $ first_impression_at : int  1775365644 1758625699 1759595981 1764531694 NA 17540...
 $ last_impression_at  : int  1785340253 1783902783 1784992694 1785306024 NA 17847...
 $ first_watch_at      : int  1775365644 1758625699 1759595981 1764531694 NA 17540...
 $ last_watch_at       : int  1785340253 1783902783 1784992694 1785306024 NA 17847...
```

```
head(views)
```

```
video_id creator_id video_length_sec published_at
1      1      27      33 1775374080
2      2     346      27 1758594709
3      3     212      18 1759578576
4      4     410      10 1764529326
5      5     279      30 1785158154
6      6     393      17 1770846715

video_url
1 https://www.tiktok.com/@sophiadavis/video/9000000000000000001
2 https://www.tiktok.com/@ethanclark_1/video/9000000000000000002
3 https://www.tiktok.com/@charlottehall/video/9000000000000000003
4 https://www.tiktok.com/@emmathomas_1/video/9000000000000000004
5 https://www.tiktok.com/@charlottelopez/video/9000000000000000005
6 https://www.tiktok.com/@harperdavis/video/9000000000000000006

impressions_n watched_n watch_rate avg_watch_seconds_when_watched
1      25      19 0.7600000      15.684211
2      50      34 0.6800000      14.852941
3      61      42 0.6885246      10.285714
4      44      32 0.7272727       5.343750
5       0       0      NA      NA
6      35      30 0.8571429      9.266667

total_watch_seconds avg_watch_share distinct_watchers_n first_impression_at
1      298      0.3612121      19      1775365644
2      505      0.3740741      33      1758625699
3      432      0.3934426      39      1759595981
4      171      0.3886364      32      1764531694
5       0      NA      0      NA
6      278      0.4672269      30      1754076599

last_impression_at first_watch_at last_watch_at
1      1785340253      1775365644      1785340253
2      1783902783      1758625699      1783902783
3      1784992694      1759595981      1784992694
4      1785306024      1764531694      1785306024
5      NA      NA      NA
6      1784757684      1754076599      1784757684
```

```
summary(views)
```

```
video_id      creator_id      video_length_sec      published_at
```

Min. :	1	Min. :	1.0	Min. :	10.00	Min. :	1.754e+09
1st Qu.:	12501	1st Qu.:	126.0	1st Qu.:	23.00	1st Qu.:	1.762e+09
Median :	25000	Median :	250.0	Median :	30.00	Median :	1.770e+09
Mean :	25000	Mean :	249.4	Mean :	30.12	Mean :	1.770e+09
3rd Qu.:	37500	3rd Qu.:	375.0	3rd Qu.:	37.00	3rd Qu.:	1.778e+09
Max. :	50000	Max. :	500.0	Max. :	60.00	Max. :	1.786e+09

video_url	impressions_n	watched_n	watch_rate
Length:50000	Min. : 0.00	Min. : 0.00	Min. : 0.0000
Class :character	1st Qu.: 12.00	1st Qu.: 9.00	1st Qu.:0.6364
Mode :character	Median : 29.00	Median : 20.00	Median :0.6956
	Mean : 39.79	Mean : 27.56	Mean :0.6941
	3rd Qu.: 56.00	3rd Qu.: 39.00	3rd Qu.:0.7534
	Max. :396.00	Max. :264.00	Max. :1.0000
			NA's :1009

avg_watch_seconds_when_watched	total_watch_seconds	avg_watch_share
Min. : 1.00	Min. : 0.0	Min. : 0.0000
1st Qu.:11.89	1st Qu.: 122.0	1st Qu.:0.3214
Median :16.29	Median : 305.0	Median :0.3732
Mean :16.35	Mean : 449.4	Mean :0.3772
3rd Qu.:20.40	3rd Qu.: 625.0	3rd Qu.:0.4271
Max. :59.00	Max. :5335.0	Max. :1.0000
NA's :1399		NA's :1009

distinct_watchers_n	first_impression_at	last_impression_at
Min. : 0.00	Min. :1.754e+09	Min. :1.754e+09
1st Qu.: 8.00	1st Qu.:1.762e+09	1st Qu.:1.785e+09
Median : 20.00	Median :1.770e+09	Median :1.785e+09
Mean : 26.41	Mean :1.770e+09	Mean :1.785e+09
3rd Qu.: 38.00	3rd Qu.:1.778e+09	3rd Qu.:1.785e+09
Max. :238.00	Max. :1.786e+09	Max. :1.786e+09
	NA's :1009	NA's :1009

first_watch_at	last_watch_at
Min. :1.754e+09	Min. :1.754e+09
1st Qu.:1.762e+09	1st Qu.:1.785e+09
Median :1.770e+09	Median :1.785e+09
Mean :1.770e+09	Mean :1.785e+09
3rd Qu.:1.778e+09	3rd Qu.:1.785e+09
Max. :1.786e+09	Max. :1.786e+09
NA's :1009	NA's :1009

```
colSums(is.na(views))
```

video_id	creator_id
0	0
video_length_sec	published_at
0	0
video_url	impressions_n
0	0
watched_n	watch_rate
0	1009
avg_watch_seconds_when_watched	total_watch_seconds
1399	0
avg_watch_share	distinct_watchers_n
1009	0
first_impression_at	last_impression_at
1009	1009
first_watch_at	last_watch_at
1009	1009

The dataset contains information about TikTok videos, including video length, impressions, watched views, creators, and watch rate.

2.3 2.3 Impression dataset overview

The impression dataset contains information about individual impressions, including user and content characteristics.

```
nrow(impressions)
```

```
[1] 94951
```

```
str(impressions)
```

```
'data.frame': 94951 obs. of 14 variables:
 $ impression_id : int 1 2 3 4 5 6 7 8 9 10 ...
 $ session_id   : int 1 1 1 1 1 1 1 1 2 3 ...
 $ user_id      : int 4 4 4 4 4 4 4 4 7 7 ...
 $ video_id     : int 3098 2313 3632 3098 3992 3901 3098 3449 429 3632 ...
 $ creator_id   : int 115 94 101 115 205 191 115 98 93 101 ...
```

```

$ shown_at          : chr  "2025-08-01T19:45:47Z" "2025-08-01T19:45:59Z" "2025-08-01T19:46:23Z" "2025-08-01T19:46:55Z" ...
$ feed_rank         : int   1 2 3 4 5 6 7 8 1 1 ...
$ source_bucket     : chr   "known" "preferred_new" "preferred_new" "known" ...
$ score_category_match : num  0.122 0.616 0.428 0.122 0.428 ...
$ score_creator_match  : num   0 0 0 0.0134 0 ...
$ score_satiation_penalty: num  0 0.008 0.044 0.096 0.086 0.144 0.134 0.124 0 0.008 ...
$ score_total        : num  -0.0673 0.3383 0.3208 -0.1381 0.2205 ...
$ mission_ids        : chr   "" "" "" "" ...
$ creator_display_name : chr  "Mason media" "Sophia Miller" "Emma daily" "Mason media" ..

```

```
head(impressions)
```

```

impression_id session_id user_id video_id creator_id shown_at
1             1          1       4       3098       115 2025-08-01T19:45:47Z
2             2          1       4       2313       94 2025-08-01T19:45:59Z
3             3          1       4       3632      101 2025-08-01T19:46:23Z
4             4          1       4       3098      115 2025-08-01T19:46:55Z
5             5          1       4       3992      205 2025-08-01T19:46:59Z
6             6          1       4       3901      191 2025-08-01T19:47:35Z

feed_rank source_bucket score_category_match score_creator_match
1          1          known           0.1220796           0.00000000
2          2 preferred_new           0.6157026           0.00000000
3          3 preferred_new           0.4280305           0.00000000
4          4          known           0.1220796           0.01335068
5          5 preferred_new           0.4280305           0.00000000
6          6 preferred_new           0.3011162           0.00000000

score_satiation_penalty score_total mission_ids creator_display_name
1              0.000 -0.06731493           Mason media
2              0.008  0.33825113           Sophia Miller
3              0.044  0.32079763           Emma daily
4              0.096 -0.13810713           Mason media
5              0.086  0.22053097           Ethan fit
6              0.144  0.00274775           Amelia hub

```

```
summary(impressions)
```

```

impression_id    session_id      user_id      video_id      creator_id
Min.   :    1  Min.   :    1  Min.   : 1.0  Min.   :    1  Min.   : 1.0
1st Qu.:23712  1st Qu.: 5868  1st Qu.:198.0  1st Qu.:1079  1st Qu.: 54.0
Median :47404  Median :12076  Median :364.0  Median :2195  Median :106.0

```

```

Mean      :47403   Mean      :12107   Mean      :362.2   Mean      :2174   Mean      :107.3
3rd Qu.:71100   3rd Qu.:18274   3rd Qu.:524.0   3rd Qu.:3263   3rd Qu.:161.0
Max.      :94790   Max.      :24684   Max.      :718.0   Max.      :4308   Max.      :215.0

  shown_at      feed_rank      source_bucket      score_category_match
Length:94951    Min.       : 1.000   Length:94951    Min.       :-2.98167
Class :character 1st Qu.: 1.000   Class :character 1st Qu.: -0.08957
Mode  :character Median : 3.000   Mode  :character Median : 0.46104
                        Mean  : 3.837                        Mean  : 0.41990
                        3rd Qu.: 5.000                        3rd Qu.: 0.95116
                        Max.   :33.000                        Max.   : 3.70116

score_creator_match score_satiation_penalty score_total
Min.       :-0.25456   Min.       :0.0000   Min.       :-2.75919
1st Qu.: 0.00000   1st Qu.:0.1490   1st Qu.: -0.40874
Median : 0.00000   Median :0.2705   Median : -0.02611
Mean    : 0.02319   Mean    :0.2981   Mean    : -0.05872
3rd Qu.: 0.03126   3rd Qu.:0.4188   3rd Qu.: 0.31608
Max.    : 0.74159   Max.    :1.4513   Max.    : 2.43924

mission_ids      creator_display_name
Length:94951     Length:94951
Class :character Class :character
Mode  :character Mode  :character

```

```
colSums(is.na(impressions))
```

```

impression_id      session_id      user_id
0                  0                  0
video_id           creator_id      shown_at
0                  0                  0
feed_rank          source_bucket    score_category_match
0                  0                  0
score_creator_match score_satiation_penalty score_total
0                  0                  0
mission_ids        creator_display_name
0                  0

```

2.4 2.4 Watch-event dataset overview

The watch-event data contain information about users' watching behaviour.

```
nrow(watch_events)
```

```
[1] 98702
```

```
str(watch_events)
```

```
'data.frame':  98702 obs. of  12 variables:
 $ watch_event_id  : int  1 2 3 4 5 6 7 8 9 10 ...
 $ impression_id   : int  1 2 3 4 5 6 7 8 9 10 ...
 $ session_id      : int  1 2 3 3 3 3 3 3 3 3 ...
 $ user_id         : int  6 8 9 9 9 9 9 9 9 9 ...
 $ video_id        : int  3224 668 3440 703 2175 2351 2217 1774 3440 3536 ...
 $ creator_id      : int  117 109 152 145 179 135 95 82 152 55 ...
 $ action          : chr  "exit_platform" "exit_platform" "skip_immediate" "watch_full" ...
 $ watch_seconds   : int  8 6 0 34 NA 17 0 38 25 0 ...
 $ started_at      : chr  "2025-08-01T13:09:01Z" "2025-08-01T21:26:57Z" "2025-08-01T17:22:41Z" "2025-08-01T17:22:46Z" ...
 $ ended_at        : chr  "2025-08-01T13:09:09Z" "2025-08-01T21:27:03Z" "2025-08-01T17:22:42Z" "2025-08-01T17:23:20Z" ...
 $ started_at_raw  : chr  "20250801 130901" "2025-08-01T21:26:57Z" "2025-08-01T17:22:41Z" "2025-08-01T17:22:46Z" ...
 $ started_at_clean: chr  "2025-08-01T13:09:01Z" "2025-08-01T21:26:57Z" "2025-08-01T17:22:41Z" "2025-08-01T17:22:46Z" ...
```

```
head(watch_events)
```

	watch_event_id	impression_id	session_id	user_id	video_id	creator_id	
1	1	1	1	1	6	3224	117
2	2	2	2	2	8	668	109
3	3	3	3	3	9	3440	152
4	4	4	4	3	9	703	145
5	5	5	5	3	9	2175	179
6	6	6	6	3	9	2351	135

	action	watch_seconds	started_at	ended_at
1	exit_platform	8	2025-08-01T13:09:01Z	2025-08-01T13:09:09Z
2	exit_platform	6	2025-08-01T21:26:57Z	2025-08-01T21:27:03Z
3	skip_immediate	0	2025-08-01T17:22:41Z	2025-08-01T17:22:42Z
4	watch_full	34	2025-08-01T17:22:46Z	2025-08-01T17:23:20Z
5	skip_immediate	NA	2025-08-01T17:23:24Z	2025-08-01T17:23:25Z
6	skip_after_partial	17	2025-08-01T17:23:29Z	2025-08-01T17:23:46Z

```

      started_at_raw      started_at_clean
1      20250801 130901 2025-08-01T13:09:01Z
2 2025-08-01T21:26:57Z 2025-08-01T21:26:57Z
3 2025-08-01T17:22:41Z 2025-08-01T17:22:41Z
4 2025-08-01T17:22:46Z 2025-08-01T17:22:46Z
5 2025-08-01T17:23:24Z 2025-08-01T17:23:24Z
6 2025-08-01T17:23:29Z 2025-08-01T17:23:29Z

```

```
summary(watch_events)
```

watch_event_id	impression_id	session_id	user_id	video_id
Min. : 1	Min. : 1	Min. : 1	Min. : 1.0	Min. : 1
1st Qu.:24676	1st Qu.:24676	1st Qu.: 6213	1st Qu.:184.0	1st Qu.:1105
Median :49352	Median :49352	Median :12678	Median :354.0	Median :2222
Mean :49352	Mean :49352	Mean :12712	Mean :359.9	Mean :2210
3rd Qu.:74027	3rd Qu.:74027	3rd Qu.:19154	3rd Qu.:541.0	3rd Qu.:3313
Max. :98702	Max. :98702	Max. :25705	Max. :725.0	Max. :4350

creator_id	action	watch_seconds	started_at
Min. : 1.0	Length:98702	Min. : 0.00	Length:98702
1st Qu.: 60.0	Class :character	1st Qu.: 1.00	Class :character
Median :112.0	Mode :character	Median : 9.00	Mode :character
Mean :111.2		Mean :12.67	
3rd Qu.:162.0		3rd Qu.:21.00	
Max. :218.0		Max. :60.00	
		NA's :7245	

ended_at	started_at_raw	started_at_clean
Length:98702	Length:98702	Length:98702
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

```
colSums(is.na(watch_events))
```

watch_event_id	impression_id	session_id	user_id
0	0	0	0
video_id	creator_id	action	watch_seconds
0	0	0	7245

started_at	ended_at	started_at_raw	started_at_clean
0	0	0	0

3 3. Descriptive Analysis

3.1 3.1 Overall video performance

We first calculate several overall performance measures for the video-view dataset.

```
overview <- views %>%
  summarise(
    videos = n(),
    creators = n_distinct(creator_id),
    avg_length = mean(video_length_sec, na.rm = TRUE),
    avg_impressions = mean(impressions_n, na.rm = TRUE),
    avg_watched = mean(watched_n, na.rm = TRUE),
    avg_watch_rate = mean(watch_rate, na.rm = TRUE),
    avg_watch_share = mean(
      watched_n / impressions_n,
      na.rm = TRUE
    ),
    avg_watch_seconds = mean(
      video_length_sec * watched_n / impressions_n,
      na.rm = TRUE
    )
  )
overview
```

	videos	creators	avg_length	avg_impressions	avg_watched	avg_watch_rate
1	50000	500	30.12118	39.79308	27.55746	0.6940614

	avg_watch_share	avg_watch_seconds
1	0.6940614	20.90968

This provides an overall picture of the number of videos and creators in the dataset, as well as average reach and watching behaviour.

3.2 3.2 Watch share

To examine the proportion of impressions that resulted in a watched view, we calculate watch share as a percentage.

```
views <- views %>%  
  mutate(  
    watched_pct = watched_n / impressions_n * 100  
  )  
  
head(views)
```

	video_id	creator_id	video_length_sec	published_at	
1	1	27	33	1775374080	
2	2	346	27	1758594709	
3	3	212	18	1759578576	
4	4	410	10	1764529326	
5	5	279	30	1785158154	
6	6	393	17	1770846715	

	video_url
1	https://www.tiktok.com/@sophiadavis/video/9000000000000000001
2	https://www.tiktok.com/@ethanclark_1/video/9000000000000000002
3	https://www.tiktok.com/@charlottehall/video/9000000000000000003
4	https://www.tiktok.com/@emmathomas_1/video/9000000000000000004
5	https://www.tiktok.com/@charlottelopez/video/9000000000000000005
6	https://www.tiktok.com/@harperdavis/video/9000000000000000006

	impressions_n	watched_n	watch_rate	avg_watch_seconds_when_watched
1	25	19	0.7600000	15.684211
2	50	34	0.6800000	14.852941
3	61	42	0.6885246	10.285714
4	44	32	0.7272727	5.343750
5	0	0	NA	NA
6	35	30	0.8571429	9.266667

	total_watch_seconds	avg_watch_share	distinct_watchers_n	first_impression_at
1	298	0.3612121	19	1775365644
2	505	0.3740741	33	1758625699
3	432	0.3934426	39	1759595981
4	171	0.3886364	32	1764531694
5	0	NA	0	NA
6	278	0.4672269	30	1754076599

	last_impression_at	first_watch_at	last_watch_at	watched_pct
1	1785340253	1775365644	1785340253	76.00000

2	1783902783	1758625699	1783902783	68.00000
3	1784992694	1759595981	1784992694	68.85246
4	1785306024	1764531694	1785306024	72.72727
5	NA	NA	NA	NaN
6	1784757684	1754076599	1784757684	85.71429

Watch share provides a measure of how much of the observed audience proceeded to watch the video.

3.3 3.3 Most viewed video

The video with the highest number of impressions is identified below.

```
topvideo <- views %>%
  filter(
    impressions_n == max(
      impressions_n,
      na.rm = TRUE
    )
  )
topvideo
```

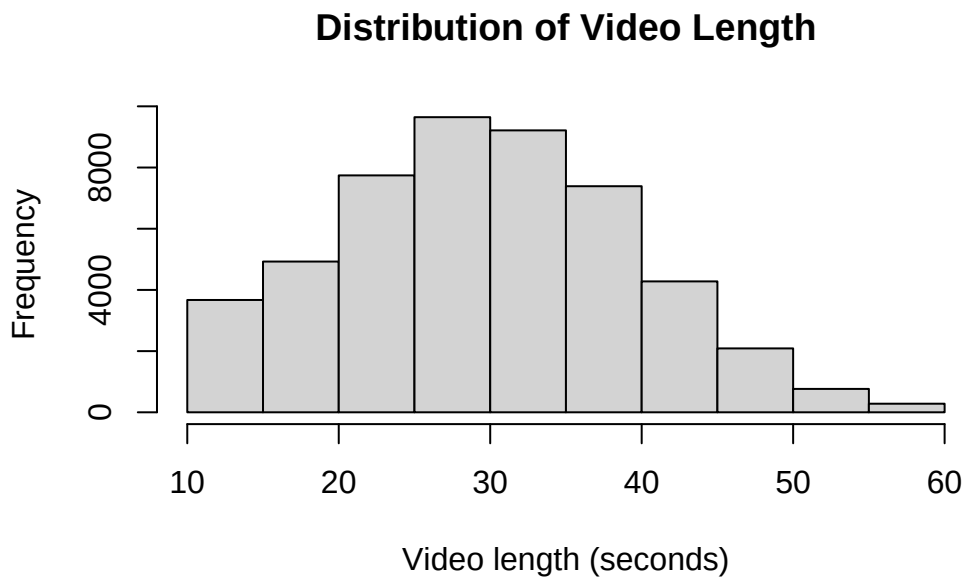
	video_id	creator_id	video_length_sec	published_at		video_url	impressions_n	
1	19968	405	20	1754087140				
1					https://www.tiktok.com/@noahwilson/video/9000000000000019968		396	
					watched_n	watch_rate	avg_watch_seconds_when_watched	total_watch_seconds
1					264	0.6666667	10.58333	2794
					avg_watch_share	distinct_watchers_n	first_impression_at	last_impression_at
1					0.3527778	238	1754007889	1785005775
					first_watch_at	last_watch_at	watched_pct	
1					1754007889	1785005775	66.66667	

This identifies the video that achieved the highest observed reach in the dataset.

3.4 3.4 Distribution of video length

The distribution of video length is visualised using a histogram.

```
hist(
  views$video_length_sec,
  main = "Distribution of Video Length",
  xlab = "Video length (seconds)"
)
```



The distribution provides an overview of the typical length of videos in the dataset and whether particularly short or long videos are common.

3.5 3.5 Videos with high watch rates

We identify videos with a watch rate above 80%.

```
goodvideos <- views %>%
  filter(
    watch_rate > 0.8
  )

nrow(goodvideos)
```

```
[1] 6601
```

This shows how many videos achieved a relatively high watch rate according to the threshold used in the analysis.

3.6 3.6 Top 10 videos by watch rate

The ten videos with the highest watch rates are displayed below.

```
top_watch_rate <- views %>%
  select(
    video_id,
    creator_id,
    impressions_n,
    watched_n,
    watch_rate
  ) %>%
  arrange(
    desc(watch_rate)
  ) %>%
  slice_head(
    n = 10
  )

top_watch_rate
```

	video_id	creator_id	impressions_n	watched_n	watch_rate
1	8	255	4	4	1
2	43	243	1	1	1
3	62	422	4	4	1
4	105	45	1	1	1
5	163	81	1	1	1
6	175	195	5	5	1
7	203	49	2	2	1
8	217	215	2	2	1
9	219	141	9	9	1
10	266	283	4	4	1

This allows the videos with the strongest observed watch rates to be compared.

3.7 Performance by creator

Video performance is also aggregated at the creator level.

```
creator_summary <- views %>%
  group_by(creator_id) %>%
  summarise(
    videos = n(),
    total_impressions = sum(
      impressions_n,
      na.rm = TRUE
    ),
    average_watch_rate = mean(
      watch_rate,
      na.rm = TRUE
    )
  ) %>%
  arrange(
    desc(total_impressions)
  )

creator_summary
```

```
# A tibble: 500 x 4
  creator_id videos total_impressions average_watch_rate
    <int>   <int>         <int>           <dbl>
1       499    283         11568             0.693
2       419    237         10519             0.682
3       294    255         10327             0.684
4       231    245         10144             0.677
5       184    253         10074             0.696
6       479    239          9936             0.688
7       281    182          9160             0.695
8        33    182          8977             0.679
9       417    244          8937             0.698
10      258    218          8933             0.702
# i 490 more rows
```

This comparison provides insight into differences in reach and average watch rate between creators.

4 4. Watch-Event Analysis

4.1 4.1 Aggregating watch events

The watch events are aggregated to the impression level. This creates a single total watch-time measure for each impression.

```
watch_summary <- watch_events %>%
  group_by(impression_id) %>%
  summarise(
    watch_seconds = sum(
      watch_seconds,
      na.rm = TRUE
    ),
    .groups = "drop"
  )

head(watch_summary)
```

```
# A tibble: 6 x 2
  impression_id watch_seconds
      <int>         <int>
1           1             8
2           2             6
3           3             0
4           4            34
5           5             0
6           6            17
```

4.2 4.2 Combining impression and watch data

The aggregated watch data are joined to the impression data.

```
reg_data <- impressions %>%
  left_join(
    watch_summary,
    by = "impression_id"
  ) %>%
  mutate(
    watch_seconds = ifelse(
      is.na(watch_seconds),
```

```

    0,
    watch_seconds
  ),
  source_bucket = as.factor(source_bucket),
  user_id = as.factor(user_id)
)

head(reg_data)

```

	impression_id	session_id	user_id	video_id	creator_id	shown_at
1	1	1	4	3098	115	2025-08-01T19:45:47Z
2	2	1	4	2313	94	2025-08-01T19:45:59Z
3	3	1	4	3632	101	2025-08-01T19:46:23Z
4	4	1	4	3098	115	2025-08-01T19:46:55Z
5	5	1	4	3992	205	2025-08-01T19:46:59Z
6	6	1	4	3901	191	2025-08-01T19:47:35Z

	feed_rank	source_bucket	score_category_match	score_creator_match
1	1	known	0.1220796	0.00000000
2	2	preferred_new	0.6157026	0.00000000
3	3	preferred_new	0.4280305	0.00000000
4	4	known	0.1220796	0.01335068
5	5	preferred_new	0.4280305	0.00000000
6	6	preferred_new	0.3011162	0.00000000

	score_satiation_penalty	score_total	mission_ids	creator_display_name
1	0.000	-0.06731493		Mason media
2	0.008	0.33825113		Sophia Miller
3	0.044	0.32079763		Emma daily
4	0.096	-0.13810713		Mason media
5	0.086	0.22053097		Ethan fit
6	0.144	0.00274775		Amelia hub

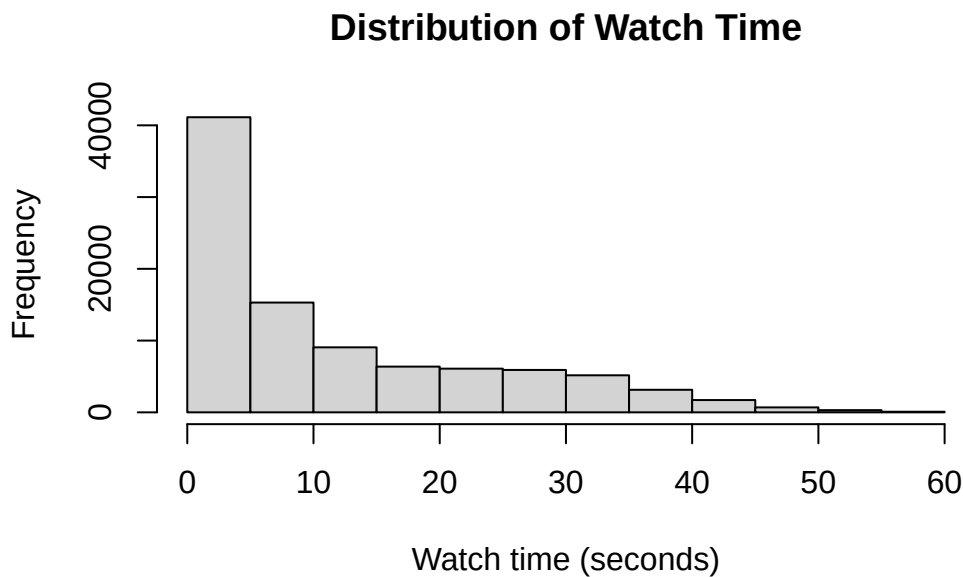
	watch_seconds
1	8
2	6
3	0
4	34
5	0
6	17

The resulting dataset contains both the characteristics of each impression and the associated total watch time.

4.3 4.3 Distribution of watch time

The distribution of watch time is shown below.

```
hist(  
  reg_data$watch_seconds,  
  main = "Distribution of Watch Time",  
  xlab = "Watch time (seconds)"  
)
```



The distribution shows how total watch time varies across impressions.

5 5. Regression Analysis

The regression analysis investigates the relationship between impression characteristics and total watch time.

The dependent variable in all three models is `watch_seconds`.

5.1 5.1 Simple regression

The first model examines the relationship between `score_total` and watch time.

```
model1 <- lm(
  watch_seconds ~ score_total,
  data = reg_data
)

summary(model1)
```

Call:

```
lm(formula = watch_seconds ~ score_total, data = reg_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-11.895	-11.712	-4.683	8.253	48.341

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	11.74126	0.04150	282.951	<2e-16 ***
score_total	-0.06256	0.07195	-0.869	0.385

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.72 on 94949 degrees of freedom

Multiple R-squared: 7.962e-06, Adjusted R-squared: -2.57e-06

F-statistic: 0.756 on 1 and 94949 DF, p-value: 0.3846

This model provides the baseline association between the total score and watch time without controlling for the other variables.

5.2 5.2 Multiple regression

The second model adds additional explanatory variables:

- `feed_rank`
- `score_category_match`
- `score_creator_match`
- `score_satiation_penalty`

- source_bucket

```
model2 <- lm(
  watch_seconds ~ score_total +
    feed_rank +
    score_category_match +
    score_creator_match +
    score_satiation_penalty +
    source_bucket,
  data = reg_data
)

summary(model2)
```

Call:

```
lm(formula = watch_seconds ~ score_total + feed_rank + score_category_match +
    score_creator_match + score_satiation_penalty + source_bucket,
    data = reg_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-12.342	-11.617	-4.496	8.229	48.656

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	11.71381	0.10634	110.159	<2e-16 ***
score_total	0.61492	0.31381	1.960	0.0501 .
feed_rank	0.01410	0.01296	1.088	0.2765
score_category_match	-0.48467	0.21142	-2.292	0.0219 *
score_creator_match	-1.61901	0.92669	-1.747	0.0806 .
score_satiation_penalty	0.53129	0.33887	1.568	0.1169
source_bucketknown	0.21236	0.14159	1.500	0.1337
source_bucketpreferred_new	0.11576	0.10182	1.137	0.2556

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.72 on 94943 degrees of freedom

Multiple R-squared: 0.0001158, Adjusted R-squared: 4.212e-05

F-statistic: 1.571 on 7 and 94943 DF, p-value: 0.1386

The multiple regression allows the association between `score_total` and watch time to be examined while controlling for the other variables included in the model.

5.3 5.3 User fixed effects

The third model additionally controls for differences between users by including `user_id` as a factor.

```
model3 <- lm(
  watch_seconds ~ score_total +
    feed_rank +
    score_category_match +
    score_creator_match +
    score_satiation_penalty +
    source_bucket +
    user_id,
  data = reg_data
)

summary(model3)
```

Call:

```
lm(formula = watch_seconds ~ score_total + feed_rank + score_category_match +
    score_creator_match + score_satiation_penalty + source_bucket +
    user_id, data = reg_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-19.499	-10.762	-4.233	8.152	50.243

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	12.196572	1.716978	7.104	1.22e-12 ***
score_total	0.569973	0.315087	1.809	0.0705 .
feed_rank	0.012567	0.013588	0.925	0.3551
score_category_match	-0.463796	0.213529	-2.172	0.0299 *
score_creator_match	-1.825899	0.948070	-1.926	0.0541 .
score_satiation_penalty	0.283652	0.391756	0.724	0.4690
source_bucketknown	0.216453	0.142468	1.519	0.1287
source_bucketpreferred_new	0.115909	0.103790	1.117	0.2641
user_id2	-1.700458	2.306626	-0.737	0.4610
user_id3	-0.426160	1.920683	-0.222	0.8244
user_id4	-0.429761	1.978222	-0.217	0.8280
user_id5	-0.029186	2.266238	-0.013	0.9897

user_id6	-1.046244	1.863535	-0.561	0.5745
user_id7	-0.648969	1.889851	-0.343	0.7313
user_id8	-0.691613	2.239768	-0.309	0.7575
user_id9	-1.523522	2.167971	-0.703	0.4822
user_id10	2.886026	2.572657	1.122	0.2619
user_id11	4.180760	2.299413	1.818	0.0690 .
user_id12	-0.384688	1.947507	-0.198	0.8434
user_id13	-1.213526	1.949911	-0.622	0.5337
user_id14	1.032684	1.977696	0.522	0.6016
user_id15	0.842848	2.216952	0.380	0.7038
user_id16	0.604698	2.414666	0.250	0.8023
user_id17	-1.312489	2.159367	-0.608	0.5433
user_id18	0.444835	1.929263	0.231	0.8176
user_id19	-1.808177	2.436965	-0.742	0.4581
user_id20	1.371938	2.177116	0.630	0.5286
user_id21	-1.880926	2.374374	-0.792	0.4283
user_id22	-0.392657	1.870752	-0.210	0.8338
user_id23	2.161512	2.176904	0.993	0.3207
user_id24	2.220039	3.208601	0.692	0.4890
user_id25	0.539435	2.145251	0.251	0.8015
user_id26	0.945310	2.041735	0.463	0.6434
user_id27	-1.475167	2.076947	-0.710	0.4775
user_id28	2.594036	2.683067	0.967	0.3336
user_id29	-0.184986	2.032706	-0.091	0.9275
user_id30	-0.925377	3.262922	-0.284	0.7767
user_id31	-1.209640	1.909318	-0.634	0.5264
user_id32	0.593129	2.556770	0.232	0.8166
user_id33	0.334764	2.206879	0.152	0.8794
user_id34	-0.751572	2.012816	-0.373	0.7089
user_id35	-0.773518	2.330304	-0.332	0.7399
user_id36	-1.090098	2.039732	-0.534	0.5930
user_id37	-1.283199	1.929177	-0.665	0.5060
user_id38	0.256332	1.877469	0.137	0.8914
user_id39	2.076870	2.143738	0.969	0.3326
user_id40	0.369004	1.981331	0.186	0.8523
user_id41	-1.784450	3.208797	-0.556	0.5781
user_id42	-2.546853	2.039896	-1.249	0.2118
user_id43	-0.355732	1.959865	-0.182	0.8560
user_id44	0.992510	2.029097	0.489	0.6247
user_id45	-2.298841	2.291893	-1.003	0.3158
user_id46	-0.464128	2.016857	-0.230	0.8180
user_id47	-1.047867	2.059465	-0.509	0.6109
user_id48	3.923338	3.027208	1.296	0.1950

user_id49	2.571183	2.414872	1.065	0.2870
user_id50	-1.109825	2.347163	-0.473	0.6363
user_id51	0.575437	2.704573	0.213	0.8315
user_id52	0.145166	1.952529	0.074	0.9407
user_id53	0.430557	2.117185	0.203	0.8389
user_id54	-1.356526	2.201730	-0.616	0.5378
user_id55	-0.099520	2.151093	-0.046	0.9631
user_id56	-3.230574	2.705045	-1.194	0.2324
user_id57	0.918754	2.425661	0.379	0.7049
user_id58	-0.902985	2.662635	-0.339	0.7345
user_id59	-0.845784	2.557298	-0.331	0.7408
user_id60	1.402831	1.964899	0.714	0.4753
user_id61	0.123641	2.092278	0.059	0.9529
user_id62	-0.399100	2.060687	-0.194	0.8464
user_id63	1.429331	1.945439	0.735	0.4625
user_id64	-0.419947	2.299212	-0.183	0.8551
user_id65	0.448098	1.972257	0.227	0.8203
user_id66	-0.189204	1.857509	-0.102	0.9189
user_id67	0.644818	2.498623	0.258	0.7964
user_id68	-1.012234	2.110334	-0.480	0.6315
user_id69	-1.740853	2.101139	-0.829	0.4074
user_id70	-1.555023	1.959906	-0.793	0.4275
user_id71	-1.360636	2.330385	-0.584	0.5593
user_id72	0.541246	2.278012	0.238	0.8122
user_id73	0.683967	1.971757	0.347	0.7287
user_id74	-0.574972	2.089834	-0.275	0.7832
user_id75	-0.252998	2.034758	-0.124	0.9010
user_id76	0.540903	1.965516	0.275	0.7832
user_id77	0.171102	2.004738	0.085	0.9320
user_id78	-0.156475	1.888144	-0.083	0.9340
user_id79	-1.389784	2.097595	-0.663	0.5076
user_id80	-1.061866	1.882370	-0.564	0.5727
user_id81	1.216648	2.801385	0.434	0.6641
user_id82	-0.502649	2.526871	-0.199	0.8423
user_id83	-1.149727	2.039085	-0.564	0.5729
user_id84	-3.912752	3.068093	-1.275	0.2022
user_id85	-0.015751	1.887098	-0.008	0.9933
user_id86	-2.679642	2.211436	-1.212	0.2256
user_id87	-0.728452	1.997756	-0.365	0.7154
user_id88	-0.860655	2.233648	-0.385	0.7000
user_id89	-1.450068	1.964925	-0.738	0.4605
user_id90	-0.324066	2.012484	-0.161	0.8721
user_id91	-0.119406	2.264523	-0.053	0.9579

user_id92	-0.737170	2.572544	-0.287	0.7745
user_id93	-0.192813	2.223497	-0.087	0.9309
user_id94	-0.015312	2.088245	-0.007	0.9941
user_id95	-0.616289	2.101107	-0.293	0.7693
user_id96	0.623060	1.857142	0.335	0.7373
user_id97	-0.464998	2.526655	-0.184	0.8540
user_id98	-0.185294	2.683437	-0.069	0.9449
user_id99	-1.240960	2.063168	-0.601	0.5475
user_id100	-1.148577	2.025669	-0.567	0.5707
user_id101	0.731500	1.944135	0.376	0.7067
user_id102	1.331561	2.953024	0.451	0.6521
user_id103	1.462532	2.088391	0.700	0.4837
user_id104	-1.574866	2.425529	-0.649	0.5162
user_id105	-0.360656	1.961220	-0.184	0.8541
user_id106	1.382046	2.355955	0.587	0.5575
user_id107	0.648211	2.201339	0.294	0.7684
user_id108	2.126574	2.315246	0.919	0.3584
user_id109	-1.849319	1.919562	-0.963	0.3353
user_id110	1.023640	1.872911	0.547	0.5847
user_id111	-3.446970	2.285159	-1.508	0.1315
user_id112	0.111880	3.068174	0.036	0.9709
user_id113	-1.090269	2.277817	-0.479	0.6322
user_id114	0.791015	1.953577	0.405	0.6855
user_id115	-0.706291	2.091290	-0.338	0.7356
user_id116	0.165844	2.121192	0.078	0.9377
user_id117	-0.789865	2.828070	-0.279	0.7800
user_id118	-0.147447	2.307270	-0.064	0.9490
user_id119	0.172180	1.947776	0.088	0.9296
user_id120	-0.600256	1.904908	-0.315	0.7527
user_id121	-1.512827	2.046095	-0.739	0.4597
user_id122	-0.391945	1.956564	-0.200	0.8412
user_id123	0.226307	2.046607	0.111	0.9120
user_id124	2.970231	2.233684	1.330	0.1836
user_id125	-1.469651	2.063691	-0.712	0.4764
user_id126	1.385116	2.186692	0.633	0.5265
user_id127	-1.205341	2.022897	-0.596	0.5513
user_id128	-3.485609	2.704575	-1.289	0.1975
user_id129	-0.150452	1.914321	-0.079	0.9374
user_id130	-1.269194	1.931598	-0.657	0.5111
user_id131	-0.230758	2.120753	-0.109	0.9134
user_id132	1.639424	2.035654	0.805	0.4206
user_id133	-0.229956	2.085231	-0.110	0.9122
user_id134	0.615284	2.128465	0.289	0.7725

user_id135	-0.513561	1.947689	-0.264	0.7920
user_id136	-0.728073	2.016668	-0.361	0.7181
user_id137	-1.246936	2.151007	-0.580	0.5621
user_id138	0.272270	2.284737	0.119	0.9051
user_id139	-1.264083	1.971773	-0.641	0.5215
user_id140	1.570416	2.292113	0.685	0.4933
user_id141	-1.227340	2.163941	-0.567	0.5706
user_id142	-0.812792	1.888652	-0.430	0.6669
user_id143	-3.284295	3.384970	-0.970	0.3319
user_id144	-0.147345	1.947474	-0.076	0.9397
user_id145	-0.097428	2.061229	-0.047	0.9623
user_id146	0.121272	1.964883	0.062	0.9508
user_id147	-1.723205	2.245691	-0.767	0.4429
user_id148	1.683402	2.556823	0.658	0.5103
user_id149	-0.876296	2.186380	-0.401	0.6886
user_id150	0.561665	1.849622	0.304	0.7614
user_id151	0.728707	1.962577	0.371	0.7104
user_id152	5.785636	2.448268	2.363	0.0181 *
user_id153	-0.492313	2.132566	-0.231	0.8174
user_id154	11.475500	6.586824	1.742	0.0815 .
user_id155	0.916301	2.186418	0.419	0.6752
user_id156	-0.800752	2.068439	-0.387	0.6987
user_id157	0.247831	2.143374	0.116	0.9079
user_id158	-0.124353	2.211436	-0.056	0.9552
user_id159	1.755399	2.330365	0.753	0.4513
user_id160	-1.511272	2.048347	-0.738	0.4606
user_id161	-1.952084	2.472961	-0.789	0.4299
user_id162	-1.578735	2.168432	-0.728	0.4666
user_id163	-0.184135	2.079998	-0.089	0.9295
user_id164	2.112215	2.124203	0.994	0.3201
user_id165	-2.730144	2.233842	-1.222	0.2216
user_id166	0.180042	2.299422	0.078	0.9376
user_id167	0.228980	2.177283	0.105	0.9162
user_id168	-0.395948	1.965469	-0.201	0.8403
user_id169	-0.982825	1.959803	-0.501	0.6160
user_id170	-2.710494	2.026032	-1.338	0.1810
user_id171	-1.111044	1.933667	-0.575	0.5656
user_id172	0.104334	2.322256	0.045	0.9642
user_id173	0.563718	2.727248	0.207	0.8362
user_id174	1.105070	2.076957	0.532	0.5947
user_id175	1.252071	2.196901	0.570	0.5687
user_id176	-0.856344	1.990341	-0.430	0.6670
user_id177	-0.936657	2.338636	-0.401	0.6888

user_id178	-2.493944	2.043879	-1.220	0.2224
user_id179	0.708366	1.994848	0.355	0.7225
user_id180	-0.284551	1.985165	-0.143	0.8860
user_id181	-0.080870	2.033695	-0.040	0.9683
user_id182	0.767135	2.151117	0.357	0.7214
user_id183	-0.750984	2.886988	-0.260	0.7948
user_id184	0.897358	1.986697	0.452	0.6515
user_id185	-1.050115	1.912439	-0.549	0.5829
user_id186	0.006733	1.936129	0.003	0.9972
user_id187	-1.400606	2.106964	-0.665	0.5062
user_id188	-0.477904	2.097565	-0.228	0.8198
user_id189	-4.513748	2.384068	-1.893	0.0583
user_id190	0.302438	2.472841	0.122	0.9027
user_id191	-2.359514	2.338719	-1.009	0.3130
user_id192	3.331965	2.365078	1.409	0.1589
user_id193	-0.627811	1.895835	-0.331	0.7405
user_id194	-0.317893	2.077129	-0.153	0.8784
user_id195	0.593993	2.285281	0.260	0.7949
user_id196	0.130684	2.048560	0.064	0.9491
user_id197	0.209470	2.006834	0.104	0.9169
user_id198	2.664024	2.007237	1.327	0.1844
user_id199	0.525528	1.943683	0.270	0.7869
user_id200	-0.452990	2.167963	-0.209	0.8345
user_id201	-1.246950	1.973700	-0.632	0.5275
user_id202	-0.364212	1.881139	-0.194	0.8465
user_id203	-0.120592	2.103933	-0.057	0.9543
user_id204	0.380444	2.058789	0.185	0.8534
user_id205	1.202787	2.306626	0.521	0.6021
user_id206	-0.691577	1.940835	-0.356	0.7216
user_id207	0.393016	1.905248	0.206	0.8366
user_id208	0.230622	1.914633	0.120	0.9041
user_id209	-1.390500	2.082747	-0.668	0.5044
user_id210	0.424208	1.891730	0.224	0.8226
user_id211	-2.676524	2.414636	-1.108	0.2677
user_id212	-1.398871	1.901795	-0.736	0.4620
user_id213	-0.100794	2.306770	-0.044	0.9651
user_id214	-0.843387	2.058069	-0.410	0.6820
user_id215	-0.424458	2.172682	-0.195	0.8451
user_id216	-2.008418	2.252005	-0.892	0.3725
user_id217	0.440157	2.110392	0.209	0.8348
user_id218	0.543846	2.415100	0.225	0.8218
user_id219	-1.502871	2.291955	-0.656	0.5120
user_id220	0.371266	2.775117	0.134	0.8936

user_id221	0.466779	1.932828	0.242	0.8092
user_id222	-0.236785	1.991367	-0.119	0.9054
user_id223	-2.053078	1.978937	-1.037	0.2995
user_id224	-0.467916	2.045166	-0.229	0.8190
user_id225	-0.325682	2.347223	-0.139	0.8896
user_id226	-0.629666	1.989766	-0.316	0.7517
user_id227	-1.788252	2.163728	-0.826	0.4085
user_id228	-2.062836	2.704742	-0.763	0.4457
user_id229	0.946186	2.034724	0.465	0.6419
user_id230	0.650451	2.330848	0.279	0.7802
user_id231	0.584771	1.876698	0.312	0.7553
user_id232	1.036748	2.774923	0.374	0.7087
user_id233	1.321185	2.251749	0.587	0.5574
user_id234	-1.551360	2.460164	-0.631	0.5283
user_id235	1.301538	2.163652	0.602	0.5475
user_id236	-1.046466	1.927008	-0.543	0.5871
user_id237	-1.869800	2.135288	-0.876	0.3812
user_id238	-1.318802	2.083788	-0.633	0.5268
user_id239	-1.757533	1.960989	-0.896	0.3701
user_id240	-1.462472	1.857815	-0.787	0.4312
user_id241	0.127934	1.890418	0.068	0.9460
user_id242	-0.077696	1.960979	-0.040	0.9684
user_id243	-1.222093	2.163767	-0.565	0.5722
user_id244	1.610995	2.036881	0.791	0.4290
user_id245	0.669784	2.299970	0.291	0.7709
user_id246	0.352842	2.526636	0.140	0.8889
user_id247	-2.635944	2.186446	-1.206	0.2280
user_id248	0.359780	2.233901	0.161	0.8721
user_id249	-1.548729	2.076660	-0.746	0.4558
user_id250	-0.483938	1.913680	-0.253	0.8004
user_id251	-0.056448	1.878923	-0.030	0.9760
user_id252	-0.890962	1.865620	-0.478	0.6330
user_id253	-2.695078	1.998276	-1.349	0.1774
user_id254	0.460833	1.879854	0.245	0.8063
user_id255	1.171218	2.284780	0.513	0.6082
user_id256	-1.327368	2.085388	-0.637	0.5244
user_id257	0.557049	2.002390	0.278	0.7809
user_id258	0.652337	1.873287	0.348	0.7277
user_id259	3.628842	5.104240	0.711	0.4771
user_id260	-0.139006	1.881291	-0.074	0.9411
user_id261	0.639047	2.240170	0.285	0.7754
user_id262	-2.041807	2.030603	-1.006	0.3147
user_id263	-0.790459	1.867259	-0.423	0.6721

user_id264	0.221355	1.868503	0.118	0.9057
user_id265	-0.625083	2.040127	-0.306	0.7593
user_id266	-1.536997	2.079654	-0.739	0.4599
user_id267	-0.387723	1.871329	-0.207	0.8359
user_id268	-3.898360	2.132066	-1.828	0.0675 .
user_id269	0.828796	1.965017	0.422	0.6732
user_id270	1.764765	2.245622	0.786	0.4319
user_id271	0.603453	2.228071	0.271	0.7865
user_id272	-0.946680	2.374692	-0.399	0.6901
user_id273	-3.126035	2.201362	-1.420	0.1556
user_id274	-1.295912	1.896916	-0.683	0.4945
user_id275	-1.484477	2.120682	-0.700	0.4839
user_id276	0.036398	3.067993	0.012	0.9905
user_id277	-3.405035	2.541490	-1.340	0.1803
user_id278	2.047942	2.307247	0.888	0.3748
user_id279	-1.400464	2.008791	-0.697	0.4857
user_id280	2.285543	2.222448	1.028	0.3038
user_id281	1.213054	1.988183	0.610	0.5418
user_id282	-0.997936	2.117117	-0.471	0.6374
user_id283	2.472775	2.091878	1.182	0.2372
user_id284	-1.239737	1.880916	-0.659	0.5098
user_id285	-0.853807	1.839266	-0.464	0.6425
user_id286	-0.033753	2.072531	-0.016	0.9870
user_id287	1.359147	2.036134	0.668	0.5044
user_id288	-2.153964	2.164570	-0.995	0.3197
user_id289	-0.323872	2.485359	-0.130	0.8963
user_id290	-1.187427	1.850883	-0.642	0.5212
user_id291	-4.472403	2.436871	-1.835	0.0665 .
user_id292	-1.002675	2.036927	-0.492	0.6225
user_id293	-0.444220	2.314399	-0.192	0.8478
user_id294	0.438081	2.055567	0.213	0.8312
user_id295	-1.662870	2.299174	-0.723	0.4695
user_id296	-1.072948	2.683065	-0.400	0.6892
user_id297	-3.953844	2.085241	-1.896	0.0579 .
user_id298	-1.142720	2.330418	-0.490	0.6239
user_id299	1.748648	2.291970	0.763	0.4455
user_id300	2.423247	2.233708	1.085	0.2780
user_id301	0.695866	1.852625	0.376	0.7072
user_id302	0.365155	1.875170	0.195	0.8456
user_id303	-0.384599	1.884424	-0.204	0.8383
user_id304	-0.692337	1.974652	-0.351	0.7259
user_id305	-0.611687	1.894789	-0.323	0.7468
user_id306	-0.423256	2.041531	-0.207	0.8358

user_id307	0.140145	1.984263	0.071	0.9437
user_id308	-0.152519	1.927738	-0.079	0.9369
user_id309	-0.611065	1.962647	-0.311	0.7555
user_id310	-1.341849	1.945570	-0.690	0.4904
user_id311	-0.466264	1.904442	-0.245	0.8066
user_id312	-1.853472	2.314342	-0.801	0.4232
user_id313	-0.807793	1.936633	-0.417	0.6766
user_id314	0.252186	1.892719	0.133	0.8940
user_id315	-0.733529	1.947346	-0.377	0.7064
user_id316	-0.072787	1.914828	-0.038	0.9697
user_id317	-1.784701	2.284906	-0.781	0.4348
user_id318	-0.120367	1.960304	-0.061	0.9510
user_id319	-0.086940	2.055821	-0.042	0.9663
user_id320	-1.477275	2.091299	-0.706	0.4799
user_id321	-0.987965	2.264549	-0.436	0.6626
user_id322	1.689475	1.938495	0.872	0.3835
user_id323	-0.336746	2.131453	-0.158	0.8745
user_id324	-3.157024	2.988943	-1.056	0.2909
user_id325	0.110726	1.965299	0.056	0.9551
user_id326	1.982808	2.394419	0.828	0.4076
user_id327	0.145184	2.039442	0.071	0.9432
user_id328	1.291034	2.227979	0.579	0.5623
user_id329	-2.055107	1.996287	-1.029	0.3033
user_id330	-0.192799	1.998337	-0.096	0.9231
user_id331	-2.336101	3.612887	-0.647	0.5179
user_id332	-1.713994	2.039101	-0.841	0.4006
user_id333	-0.064713	1.898575	-0.034	0.9728
user_id334	-0.385972	1.952167	-0.198	0.8433
user_id335	-0.354578	2.228038	-0.159	0.8736
user_id336	-1.955012	2.079500	-0.940	0.3472
user_id337	-2.815264	2.322564	-1.212	0.2255
user_id338	-3.026760	1.949545	-1.553	0.1205
user_id339	0.947748	2.117407	0.448	0.6544
user_id340	-1.888045	2.019377	-0.935	0.3498
user_id341	0.150263	1.970923	0.076	0.9392
user_id342	-1.539424	2.384094	-0.646	0.5185
user_id343	-0.979422	1.929885	-0.508	0.6118
user_id344	0.586204	2.031108	0.289	0.7729
user_id345	-0.348222	2.147030	-0.162	0.8712
user_id346	-0.096640	2.085478	-0.046	0.9630
user_id347	-1.250998	2.121321	-0.590	0.5554
user_id348	-0.286556	2.121317	-0.135	0.8925
user_id349	0.320737	2.041667	0.157	0.8752

user_id350	-2.029464	1.969374	-1.031	0.3028
user_id351	-1.548014	2.159649	-0.717	0.4735
user_id352	1.420997	1.956546	0.726	0.4677
user_id353	-0.286443	1.972065	-0.145	0.8845
user_id354	1.197843	1.873792	0.639	0.5227
user_id355	0.055471	2.365445	0.023	0.9813
user_id356	-1.116989	2.384233	-0.468	0.6394
user_id357	-1.092166	1.906131	-0.573	0.5667
user_id358	1.814481	3.612778	0.502	0.6155
user_id359	-0.385694	2.068642	-0.186	0.8521
user_id360	2.078027	2.060209	1.009	0.3131
user_id361	-0.879490	2.001222	-0.439	0.6603
user_id362	-0.011029	1.901653	-0.006	0.9954
user_id363	-1.259153	1.942250	-0.648	0.5168
user_id364	-0.760824	1.901090	-0.400	0.6890
user_id365	-3.584228	2.228175	-1.609	0.1077
user_id366	-1.096888	1.989329	-0.551	0.5814
user_id367	-0.835197	1.967883	-0.424	0.6713
user_id368	0.541660	1.897613	0.285	0.7753
user_id369	-1.233608	2.124141	-0.581	0.5614
user_id370	-0.204687	1.951140	-0.105	0.9165
user_id371	-0.625939	2.005004	-0.312	0.7549
user_id372	-0.592139	1.974786	-0.300	0.7643
user_id373	1.526347	2.264480	0.674	0.5003
user_id374	0.157013	2.011279	0.078	0.9378
user_id375	-2.398749	2.177202	-1.102	0.2706
user_id376	1.039326	2.186373	0.475	0.6345
user_id377	-0.070520	2.110541	-0.033	0.9733
user_id378	-0.796647	2.083067	-0.382	0.7021
user_id379	-1.619341	2.233747	-0.725	0.4685
user_id380	-0.678714	1.973713	-0.344	0.7309
user_id381	-0.503166	1.983701	-0.254	0.7998
user_id382	-0.984464	1.931840	-0.510	0.6103
user_id383	-1.033129	2.251778	-0.459	0.6464
user_id384	-2.853137	2.100607	-1.358	0.1744
user_id385	-1.356587	2.264543	-0.599	0.5491
user_id386	0.511496	1.911895	0.268	0.7891
user_id387	1.499069	2.201757	0.681	0.4960
user_id388	-2.253908	2.094381	-1.076	0.2819
user_id389	-0.787988	2.306830	-0.342	0.7327
user_id390	-1.642659	2.314365	-0.710	0.4779
user_id391	-1.110398	1.932025	-0.575	0.5655
user_id392	0.459826	1.953277	0.235	0.8139

user_id393	0.062005	2.008805	0.031	0.9754
user_id394	0.734271	1.893345	0.388	0.6982
user_id395	-0.382546	2.131714	-0.179	0.8576
user_id396	0.363603	1.882422	0.193	0.8468
user_id397	-1.584827	2.228100	-0.711	0.4769
user_id398	-0.553080	1.869207	-0.296	0.7673
user_id399	1.514343	2.384022	0.635	0.5253
user_id400	-1.434777	2.117406	-0.678	0.4980
user_id401	-0.162498	1.898380	-0.086	0.9318
user_id402	0.051026	2.073851	0.025	0.9804
user_id403	-0.866536	1.909146	-0.454	0.6499
user_id404	-1.389198	1.982122	-0.701	0.4834
user_id405	-0.675676	1.893805	-0.357	0.7213
user_id406	-1.405088	2.002122	-0.702	0.4828
user_id407	-1.882820	2.014359	-0.935	0.3499
user_id408	-0.280990	1.978129	-0.142	0.8870
user_id409	1.000419	1.926514	0.519	0.6036
user_id410	0.255273	2.007172	0.127	0.8988
user_id411	-0.348069	2.091251	-0.166	0.8678
user_id412	-0.390516	2.065880	-0.189	0.8501
user_id413	0.309661	2.228110	0.139	0.8895
user_id414	-1.418265	1.886117	-0.752	0.4521
user_id415	0.651745	3.068117	0.212	0.8318
user_id416	-1.429488	2.245657	-0.637	0.5244
user_id417	0.456326	2.143354	0.213	0.8314
user_id418	-1.329172	2.251817	-0.590	0.5550
user_id419	-0.664634	1.945841	-0.342	0.7327
user_id420	-1.295263	2.030431	-0.638	0.5235
user_id421	-1.436026	1.900899	-0.755	0.4500
user_id422	0.466771	2.000661	0.233	0.8155
user_id423	-0.015482	1.928350	-0.008	0.9936
user_id424	0.970129	1.936096	0.501	0.6163
user_id425	0.564242	1.997286	0.283	0.7776
user_id426	0.688026	2.233617	0.308	0.7581
user_id427	-0.420581	1.872578	-0.225	0.8223
user_id428	-1.153650	1.985369	-0.581	0.5612
user_id429	-0.061542	1.985462	-0.031	0.9753
user_id430	-0.471832	2.277935	-0.207	0.8359
user_id431	1.991459	2.556928	0.779	0.4361
user_id432	2.727786	2.233757	1.221	0.2220
user_id433	-0.727484	1.977637	-0.368	0.7130
user_id434	-1.531085	2.177453	-0.703	0.4820
user_id435	-0.497636	1.927232	-0.258	0.7962

user_id436	-0.265734	2.151447	-0.124	0.9017
user_id437	0.029310	1.947229	0.015	0.9880
user_id438	-1.458683	1.845963	-0.790	0.4294
user_id439	0.793016	2.070144	0.383	0.7017
user_id440	-3.864429	4.372748	-0.884	0.3768
user_id441	1.560860	1.889371	0.826	0.4087
user_id442	-0.358354	1.897359	-0.189	0.8502
user_id443	-3.617204	2.355993	-1.535	0.1247
user_id444	-1.329212	2.024289	-0.657	0.5114
user_id445	1.190544	1.871470	0.636	0.5247
user_id446	-0.902191	1.891187	-0.477	0.6333
user_id447	-0.929381	1.964655	-0.473	0.6362
user_id448	-1.965077	2.001862	-0.982	0.3263
user_id449	-0.510456	2.066610	-0.247	0.8049
user_id450	-0.240421	2.039712	-0.118	0.9062
user_id451	-1.122408	2.003251	-0.560	0.5753
user_id452	0.211319	2.182476	0.097	0.9229
user_id453	0.127090	1.856620	0.068	0.9454
user_id454	-0.370634	2.526577	-0.147	0.8834
user_id455	-1.534331	2.384047	-0.644	0.5198
user_id456	-0.660551	1.953647	-0.338	0.7353
user_id457	-0.427838	2.020229	-0.212	0.8323
user_id458	-1.556353	2.113779	-0.736	0.4616
user_id459	1.357833	2.750447	0.494	0.6215
user_id460	-0.786453	2.022444	-0.389	0.6974
user_id461	-1.218097	2.063952	-0.590	0.5551
user_id462	0.927602	1.892769	0.490	0.6241
user_id463	0.135377	2.330712	0.058	0.9537
user_id464	0.533650	2.072168	0.258	0.7968
user_id465	0.293914	2.606503	0.113	0.9102
user_id466	-0.184484	1.907083	-0.097	0.9229
user_id467	0.085211	2.800952	0.030	0.9757
user_id468	0.226178	2.135518	0.106	0.9157
user_id469	-2.045804	4.201153	-0.487	0.6263
user_id470	-1.718961	1.929765	-0.891	0.3731
user_id471	-1.166246	2.058773	-0.566	0.5711
user_id472	-1.050943	2.073946	-0.507	0.6123
user_id473	-0.809193	1.959667	-0.413	0.6797
user_id474	-1.685677	1.917132	-0.879	0.3793
user_id475	-1.575319	2.049112	-0.769	0.4420
user_id476	0.242947	1.887419	0.129	0.8976
user_id477	-0.262721	2.856539	-0.092	0.9267
user_id478	0.350706	1.999440	0.175	0.8608

user_id479	-0.198279	1.940908	-0.102	0.9186
user_id480	-0.202302	1.949256	-0.104	0.9173
user_id481	-0.817457	1.914229	-0.427	0.6693
user_id482	-0.395899	1.886190	-0.210	0.8338
user_id483	-1.792657	1.954056	-0.917	0.3589
user_id484	0.744228	2.314358	0.322	0.7478
user_id485	-1.683786	1.891673	-0.890	0.3734
user_id486	-2.862639	3.384791	-0.846	0.3977
user_id487	-0.399506	2.048309	-0.195	0.8454
user_id488	0.403167	2.071652	0.195	0.8457
user_id489	-0.387874	2.222466	-0.175	0.8615
user_id490	-0.260752	1.900998	-0.137	0.8909
user_id491	-2.034333	1.954977	-1.041	0.2981
user_id492	-0.885312	2.704566	-0.327	0.7434
user_id493	2.725063	2.643906	1.031	0.3027
user_id494	-0.865897	1.862658	-0.465	0.6420
user_id495	1.862073	2.704490	0.689	0.4911
user_id496	-1.744020	1.939955	-0.899	0.3687
user_id497	0.571548	2.148459	0.266	0.7902
user_id498	0.481728	1.919371	0.251	0.8018
user_id499	-1.128201	1.867838	-0.604	0.5458
user_id500	-2.284421	1.955271	-1.168	0.2427
user_id501	-0.436730	1.893602	-0.231	0.8176
user_id502	1.127542	2.002525	0.563	0.5734
user_id503	0.126215	1.847218	0.068	0.9455
user_id504	-2.188791	1.901727	-1.151	0.2498
user_id505	0.244978	2.365161	0.104	0.9175
user_id506	-2.635994	2.774996	-0.950	0.3422
user_id507	-1.416558	2.201046	-0.644	0.5198
user_id508	-1.056518	1.981436	-0.533	0.5939
user_id509	2.227999	1.948864	1.143	0.2529
user_id510	3.043177	3.068110	0.992	0.3213
user_id511	0.447782	2.234356	0.200	0.8412
user_id512	-1.265146	2.008896	-0.630	0.5288
user_id513	2.477945	2.048362	1.210	0.2264
user_id514	-0.115599	1.945483	-0.059	0.9526
user_id515	0.219592	1.863796	0.118	0.9062
user_id516	0.170256	2.074172	0.082	0.9346
user_id517	-1.006533	2.229222	-0.452	0.6516
user_id518	-2.112622	2.460201	-0.859	0.3905
user_id519	2.974833	2.952780	1.007	0.3137
user_id520	0.089206	1.892247	0.047	0.9624
user_id521	-2.022913	2.186427	-0.925	0.3549

user_id522	-0.890540	2.292483	-0.388	0.6977
user_id523	-0.544654	2.163682	-0.252	0.8013
user_id524	-1.332010	1.858599	-0.717	0.4736
user_id525	-2.653381	2.186340	-1.214	0.2249
user_id526	-0.976906	2.135351	-0.457	0.6473
user_id527	-0.549554	2.053105	-0.268	0.7890
user_id528	-0.341678	2.284752	-0.150	0.8811
user_id529	-0.942270	1.988173	-0.474	0.6355
user_id530	-1.310983	2.100819	-0.624	0.5326
user_id531	-0.415541	2.085224	-0.199	0.8420
user_id532	1.748310	2.662734	0.657	0.5114
user_id533	-1.744601	2.005305	-0.870	0.3843
user_id534	0.345930	2.827978	0.122	0.9026
user_id535	-0.193050	2.024127	-0.095	0.9240
user_id536	-1.483075	2.472619	-0.600	0.5486
user_id537	1.324357	2.365157	0.560	0.5755
user_id538	-3.096858	3.529572	-0.877	0.3803
user_id539	-3.518644	2.726820	-1.290	0.1969
user_id540	1.123461	1.986884	0.565	0.5718
user_id541	-1.709606	1.998669	-0.855	0.3923
user_id542	2.663621	2.589326	1.029	0.3036
user_id543	-2.175912	2.356290	-0.923	0.3558
user_id544	0.413926	2.063723	0.201	0.8410
user_id545	0.546380	2.222669	0.246	0.8058
user_id546	2.465763	2.163544	1.140	0.2544
user_id547	1.347253	2.216775	0.608	0.5434
user_id548	-0.636194	2.206458	-0.288	0.7731
user_id549	-1.179578	1.979121	-0.596	0.5512
user_id550	-1.246646	2.271172	-0.549	0.5831
user_id551	0.361700	1.938128	0.187	0.8520
user_id552	0.391678	2.046176	0.191	0.8482
user_id553	0.577852	1.934686	0.299	0.7652
user_id554	-0.865303	2.142987	-0.404	0.6864
user_id555	-1.470424	2.018354	-0.729	0.4663
user_id556	0.477073	2.089454	0.228	0.8194
user_id557	0.042458	2.683173	0.016	0.9874
user_id558	-1.266462	2.181723	-0.580	0.5616
user_id559	-0.217048	2.258185	-0.096	0.9234
user_id560	0.307994	1.892481	0.163	0.8707
user_id561	0.301135	2.222667	0.135	0.8922
user_id562	-2.188044	2.143682	-1.021	0.3074
user_id563	-0.355923	2.003859	-0.178	0.8590
user_id564	-3.017137	2.436828	-1.238	0.2157

user_id565	-0.930660	2.094284	-0.444	0.6568
user_id566	-1.581411	2.014343	-0.785	0.4324
user_id567	-0.854517	2.017192	-0.424	0.6718
user_id568	-0.865498	2.024116	-0.428	0.6689
user_id569	1.122397	2.104258	0.533	0.5938
user_id570	-0.062018	2.058570	-0.030	0.9760
user_id571	-1.092028	2.041519	-0.535	0.5927
user_id572	0.186611	1.980715	0.094	0.9249
user_id573	-2.131589	2.113684	-1.008	0.3132
user_id574	-3.973460	2.338679	-1.699	0.0893
user_id575	2.979430	2.394209	1.244	0.2133
user_id576	-1.939638	2.228136	-0.871	0.3840
user_id577	-0.033290	2.022529	-0.016	0.9869
user_id578	-12.149991	9.155809	-1.327	0.1845
user_id579	-2.382916	2.196509	-1.085	0.2780
user_id580	-3.468485	2.856925	-1.214	0.2247
user_id581	-0.515900	1.888796	-0.273	0.7847
user_id582	0.273656	2.285396	0.120	0.9047
user_id583	0.147578	2.113692	0.070	0.9443
user_id584	-0.394579	2.107270	-0.187	0.8515
user_id585	-1.645292	1.846625	-0.891	0.3729
user_id586	-1.552083	2.472647	-0.628	0.5302
user_id587	-1.457496	2.007152	-0.726	0.4677
user_id588	-1.507795	2.068848	-0.729	0.4661
user_id589	0.905798	2.919177	0.310	0.7563
user_id590	-2.392192	1.899394	-1.259	0.2079
user_id591	-4.790805	3.208648	-1.493	0.1354
user_id592	-1.140549	1.909929	-0.597	0.5504
user_id593	-3.004042	3.208693	-0.936	0.3492
user_id594	-3.305136	2.176876	-1.518	0.1289
user_id595	0.426886	2.322241	0.184	0.8542
user_id596	-1.303049	1.957350	-0.666	0.5056
user_id597	0.517006	1.932580	0.268	0.7891
user_id598	1.445230	2.330400	0.620	0.5352
user_id599	-0.183033	2.049555	-0.089	0.9288
user_id600	-1.739086	2.041592	-0.852	0.3943
user_id601	1.261753	2.258281	0.559	0.5764
user_id602	-0.388012	2.112278	-0.184	0.8543
user_id603	-0.234638	1.963381	-0.120	0.9049
user_id604	-1.921219	2.098014	-0.916	0.3598
user_id605	-0.011714	2.383985	-0.005	0.9961
user_id606	0.406178	2.163747	0.188	0.8511
user_id607	3.033303	5.104128	0.594	0.5523

user_id608	-0.848094	1.865105	-0.455	0.6493
user_id609	-0.563322	2.054032	-0.274	0.7839
user_id610	-0.503253	2.007099	-0.251	0.8020
user_id611	0.380291	2.041363	0.186	0.8522
user_id612	-0.632901	2.088279	-0.303	0.7618
user_id613	0.940587	2.182262	0.431	0.6665
user_id614	0.271181	2.186337	0.124	0.9013
user_id615	-0.166636	2.005689	-0.083	0.9338
user_id616	-2.289077	2.572587	-0.890	0.3736
user_id617	-0.427324	2.356054	-0.181	0.8561
user_id618	1.252273	2.347292	0.533	0.5937
user_id619	-0.283326	2.245541	-0.126	0.8996
user_id620	1.012599	2.394371	0.423	0.6724
user_id621	0.643532	2.485501	0.259	0.7957
user_id622	0.670753	2.233674	0.300	0.7640
user_id623	-1.636493	2.191121	-0.747	0.4551
user_id624	-0.174139	1.969169	-0.088	0.9295
user_id625	-1.574783	2.039621	-0.772	0.4401
user_id626	-1.452593	1.928121	-0.753	0.4512
user_id627	-0.346978	1.887812	-0.184	0.8542
user_id628	0.058792	2.589340	0.023	0.9819
user_id629	0.403548	1.954960	0.206	0.8365
user_id630	0.809021	1.887385	0.429	0.6682
user_id631	-0.649368	2.404549	-0.270	0.7871
user_id632	-0.526835	2.206268	-0.239	0.8113
user_id633	-1.684519	2.953044	-0.570	0.5684
user_id634	-0.146324	1.926584	-0.076	0.9395
user_id635	0.901518	2.572567	0.350	0.7260
user_id636	0.374584	2.105742	0.178	0.8588
user_id637	-0.382167	1.926676	-0.198	0.8428
user_id638	-0.442170	1.872722	-0.236	0.8133
user_id639	0.195202	2.113889	0.092	0.9264
user_id640	7.128079	3.027162	2.355	0.0185 *
user_id641	-1.252062	1.964407	-0.637	0.5239
user_id642	0.865179	2.448213	0.353	0.7238
user_id643	0.618085	2.357058	0.262	0.7931
user_id644	-0.279968	1.975305	-0.142	0.8873
user_id645	0.694027	2.042430	0.340	0.7340
user_id646	0.442114	2.006649	0.220	0.8256
user_id647	0.506108	2.044334	0.248	0.8045
user_id648	-0.964645	2.086210	-0.462	0.6438
user_id649	-1.133085	1.971771	-0.575	0.5655
user_id650	-0.822156	1.916747	-0.429	0.6680

user_id651	0.897427	2.117319	0.424	0.6717
user_id652	1.450730	2.314344	0.627	0.5308
user_id653	-2.022628	2.026218	-0.998	0.3182
user_id654	-0.803164	1.934300	-0.415	0.6780
user_id655	-1.789149	2.016174	-0.887	0.3749
user_id656	0.540433	1.902884	0.284	0.7764
user_id657	-0.145519	1.961150	-0.074	0.9409
user_id658	-1.246461	2.010620	-0.620	0.5353
user_id659	0.093824	2.101071	0.045	0.9644
user_id660	1.884867	2.035112	0.926	0.3544
user_id661	-1.964758	1.916680	-1.025	0.3053
user_id662	-1.027153	1.959727	-0.524	0.6002
user_id663	-1.104675	1.955122	-0.565	0.5721
user_id664	-0.524904	2.117305	-0.248	0.8042
user_id665	-1.381846	2.201125	-0.628	0.5301
user_id666	-1.247697	2.196156	-0.568	0.5699
user_id667	-0.338655	2.043633	-0.166	0.8684
user_id668	-1.133189	1.922923	-0.589	0.5557
user_id669	1.320164	2.113693	0.625	0.5323
user_id670	0.286252	1.879266	0.152	0.8789
user_id671	-1.401110	1.917206	-0.731	0.4649
user_id672	-1.960820	2.046107	-0.958	0.3379
user_id673	-2.582276	2.886832	-0.895	0.3711
user_id674	-0.406294	2.953353	-0.138	0.8906
user_id675	-3.517310	2.953035	-1.191	0.2336
user_id676	-0.805675	2.094402	-0.385	0.7005
user_id677	0.041611	1.902127	0.022	0.9825
user_id678	-2.442772	3.158691	-0.773	0.4393
user_id679	-0.736404	2.163819	-0.340	0.7336
user_id680	-1.351138	2.035054	-0.664	0.5067
user_id681	0.443455	2.526793	0.176	0.8607
user_id682	0.277220	2.002500	0.138	0.8899
user_id683	-0.002787	2.042527	-0.001	0.9989
user_id684	0.314959	2.063433	0.153	0.8787
user_id685	-0.974709	2.774799	-0.351	0.7254
user_id686	-1.726739	2.103757	-0.821	0.4118
user_id687	-0.338128	2.322148	-0.146	0.8842
user_id688	-1.091270	2.085781	-0.523	0.6008
user_id689	0.018593	2.065985	0.009	0.9928
user_id690	-0.340265	2.025211	-0.168	0.8666
user_id691	0.161913	2.258199	0.072	0.9428
user_id692	-0.343996	1.891892	-0.182	0.8557
user_id693	-0.586328	1.972650	-0.297	0.7663

user_id694	-0.856559	2.128752	-0.402	0.6874
user_id695	-0.024408	2.556990	-0.010	0.9924
user_id696	-1.594313	2.143048	-0.744	0.4569
user_id697	-0.044314	2.726760	-0.016	0.9870
user_id698	0.440244	2.222274	0.198	0.8430
user_id699	0.058170	1.895176	0.031	0.9755
user_id700	-4.755249	3.262792	-1.457	0.1450
user_id701	-1.938138	1.900935	-1.020	0.3079
user_id702	-0.363340	1.850428	-0.196	0.8443
user_id703	1.014106	2.030509	0.499	0.6175
user_id704	0.111704	2.152333	0.052	0.9586
user_id705	-2.043752	1.976953	-1.034	0.3012
user_id706	-3.190055	1.988431	-1.604	0.1086
user_id707	-1.089693	2.076621	-0.525	0.5998
user_id708	0.885763	2.135269	0.415	0.6783
user_id709	4.463149	2.827951	1.578	0.1145
user_id710	0.493165	2.092807	0.236	0.8137
user_id711	-1.302704	1.874020	-0.695	0.4870
user_id712	-1.553043	1.963515	-0.791	0.4290
user_id713	-0.226683	2.239999	-0.101	0.9194
user_id714	0.478117	2.120636	0.225	0.8216
user_id715	-2.346347	2.291865	-1.024	0.3059
user_id716	-0.299363	1.894789	-0.158	0.8745
user_id717	-0.141225	2.314259	-0.061	0.9513
user_id718	0.907744	2.059308	0.441	0.6594

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 12.72 on 94226 degrees of freedom

Multiple R-squared: 0.007822, Adjusted R-squared: 0.0001985

F-statistic: 1.026 on 724 and 94226 DF, p-value: 0.3062

Including `user_id` controls for systematic differences between users that could otherwise be associated with differences in watch time.

6 6. Conclusions

The analysis looked at two main things: **how TikTok videos performed** and **which factors were related to how long users watched a video**.

The descriptive analysis showed that video performance was very different across videos and creators. Some videos received many more impressions than others, and some videos had higher watch rates. This shows that not every video performed in the same way.

6.0.1 What did the regression analysis test?

The main question was:

Is a higher TikTok recommendation score associated with a longer watch time?

To test this, the watch events were first combined for each impression. This gave the total number of seconds that the user watched after that impression.

Three regression models were then tested.

6.0.2 Model 1: Recommendation score only

The first model looked only at the relationship between `score_total` and `watch_seconds`.

The result was **not statistically significant** ($p = 0.385$). Therefore, this model did **not provide sufficient evidence of a relationship between the recommendation score and watch time**.

In simple terms:

A higher recommendation score was not clearly associated with users watching the video for longer.

6.0.3 Model 2: Adding other recommendation factors

The second model added:

- feed rank
- category matching
- creator matching
- satiation penalty
- source

After adding these variables, `score_total` had a positive coefficient of **0.61**. However, its p-value was **0.0501**, which is just above the conventional 0.05 significance level.

Therefore, this model still does **not provide strong evidence** that `score_total` is related to watch time.

The overall model was also **not statistically significant** ($p = 0.139$).

One variable, `score_category_match`, was statistically significant ($p = 0.022$). Its coefficient was **-0.48**, meaning that higher values of category matching were associated with slightly lower watch time in this dataset.

This is an **association only**. It does not mean that increasing category matching would cause users to watch less.

6.0.4 Model 3: Also accounting for differences between users

The third model added `user_id` to account for differences between users.

After doing this, the coefficient for `score_total` was **0.57**, but its p-value was **0.071**.

Because this is above 0.05, the result is **not statistically significant at the 5% level**.

The relationship between `score_total` and watch time therefore remains **uncertain**.

The relationship for `score_category_match` remained statistically significant ($p = 0.030$) and negative (**-0.46**).

6.0.5 What does this mean overall?

The main finding is that the analysis **does not provide strong evidence that the overall recommendation score (`score_total`) is related to how long users watch a video**.

Although `score_total` had a positive coefficient in Models 2 and 3, the statistical evidence was not strong enough to conclude that there is a clear relationship.

The analysis did find a statistically significant negative association between `score_category_match` and watch time in Models 2 and 3. However, this should be interpreted carefully because the analysis shows an association, **not a causal effect**.

The regression models also explained only a very small part of the differences in watch time. The final model had an R^2 of **0.0078**, meaning that it explained approximately **0.8% of the variation in watch time**. :contentReferenceoacite:0

This means that most of the differences in watch time were **not explained by the variables included in these regression models**. Other factors that were not included in the dataset may play a much larger role.

Overall, the analysis shows that **video and user behaviour varies considerably, but the variables tested in the regression provide only limited information about why users watch some impressions for longer than others**.

7 7. Limitations

Several limitations should be considered when interpreting the results.

First, the analysis is based on observational data. Therefore, the regression models identify associations but cannot by themselves establish causal relationships.

Second, the models only include the variables available in the datasets. Other characteristics of users, videos, or the recommendation system may influence watch behaviour but are not necessarily captured in the models.

Third, the results of the regression analysis depend on the specification of the models. Different variables, transformations, or model specifications could lead to different estimates.

Finally, the user fixed-effects specification controls for differences between users, but it does not eliminate all potential sources of omitted-variable bias or establish causality.

8 8. Overall Conclusion

Overall, the analysis provides a descriptive overview of TikTok video performance and an examination of factors associated with user watch time. The descriptive analysis highlights differences in reach and watch behaviour across videos and creators, while the regression analysis provides a more structured examination of the relationship between impression characteristics and watch time.

The findings should therefore be interpreted as evidence of patterns and associations in the observed TikTok data rather than as evidence of causal effects.